

REPORT

Challenges Encountered & Parameter Contributions

1. Challenges Encountered

Basically I did not encounter several challenges because the two core functions inside of my `def canny_edge_detection(image, sigma, low_threshold, high_threshold)` were already built-in however in theory I reckon that non-maximum suppression sounds easy to understand (comparing a pixel with its neighbors along the gradient direction) but implementing this particular concept requires handling

- Sub-pixel interpolation when the gradient direction isn't exactly horizontal, vertical, or diagonal.
- Boundary conditions (image edges).
- Quantization of angles into 4 or 8 directions without losing precision.

Besides, I believe that sigma is a sensitive parameter and can easily effect the final result so tuning it manually without visualization is very hard.

2. Parameter Contributions (Observations from Tuning)

Sigma: Low σ (e.g., 0.5–1.0) → detects fine edges but lots of noise. High σ (e.g., 2.0–3.0) → smoother, thicker edges, may lose small details.

Recommendation: Start with 1.0–1.5 for natural images.

Low threshold: Too low → many false edges (noise). Too high → broken edges.

Recommendation: ~30–80 for 8 bit images.

High threshold: Should be 2–3× low threshold. Too high → only strongest edges survive. Too close to low → no hysteresis effect.

Recommendation: ratio 2:1 to 3:1 (e.g., low=50, high=150).

